

Practice Data Wrangling

Overview

This is a low-stakes practice assignment designed to help you build confidence with the data wrangling skills you learned in LA-4: reading data into R, using the pipe operator (`|>`), and using `filter()` and `summarise()` to answer questions about a dataset. You will use the same `broadway.csv` data as in LA-4.

Tip

If you get stuck, it can help to open your completed LA-4 script side-by-side with this assignment — the pattern of code you used there will work here, too. You can also reopen the `broadway-codebook.csv` file to remind yourself what each variable measures.

Instructions

Tip

It is often helpful to read the assignment in its entirety before starting to work on it.

- 1) Set up your R script for this assignment (name, uNID, date, assignment number). Install (if necessary) and load the packages below.
 - `tidyverse`
 - `summarytools`
 - `rstatix`
- 2) Read `broadway.csv` into R like you did in LA-4 using the `read_csv()` function.
- 3) Use the pipe operator (`|>`), and the `filter()` and `summarise()` functions to answer the following questions:
 - a) What is the minimum number of people who attended *Miss Saigon*?
 - b) What is the maximum number of people who attended *Miss Saigon*?
 - c) What is the mean number of people who attended *Miss Saigon*?

Note

This follows the same pattern you used in LA-4 to find the mean, minimum, and maximum attendance for *Mamma Mia!* and *The Lion King*. In general, that pattern looks like this:

```
df |>
  filter(variable == "value") |>
  summarise(name1 = mean(variable, na.rm = TRUE),
            name2 = min(variable, na.rm = TRUE),
            name3 = max(variable, na.rm = TRUE))
```

Replace `df` with the name of your data frame, `variable` with the relevant column name, and `"value"` with

the show you want to filter for.

- 4) Let's run a frequency distribution to see how many of each type of show (i.e., musicals, plays, specials) are in our dataset. To do this, we use the `freq()` function, which is included in the `summarytools` package. Use the frequency table that is output in the Console to determine how many shows were plays.

i Note

A **frequency distribution** (or frequency table) shows how many cases (rows) fall into each category of a variable. For example, running `freq(bway$Show.Type)` will produce a table listing each category (e.g., "Musical," "Play," "Special") along with the count and percentage of rows in the dataset that fall into that category. Find the row for "Play" and use the count (labeled **Freq**) in that row to answer the question. The template for this function is: `freq(dataframe$variable)`.

- 5) How many plays, musicals, and specials were performed at the Brooks Atkinson Theater in New York? You may need to examine a frequency distribution of the names of the theatres in the dataset to determine how to filter your data so that you only examine one specific theatre.

💡 Tip

The variable that contains theatre names is called `Show.Theater`. Running `freq(bway$Show.Theater)` will show you every theatre name exactly as it is spelled and capitalized in the dataset — this matters, because R will not recognize "brooks atkinson" as a match for "Brooks Atkinson." Once you know the exact spelling, `filter()` for that theatre and then run `freq()` again on `Show.Type` to count the plays, musicals, and specials within that theatre.

- 6) How many musicals and plays were there at the Stephen Sondheim theatre in 2011?

i Note

This question asks you to filter on *two* conditions at once: theatre name **and** year. In LA-4 (Q10), you did this by chaining two separate `filter()` steps in your pipe, like this:

```
df |>
  filter(variable1 == "value1") |>
  filter(variable2 == value2) |>
  ...
```

You can also combine both conditions inside a *single* `filter()` step using the `&` symbol, which means "and":

```
df |>
  filter(variable1 == "value1" & variable2 == value2) |>
  ...
```

Both approaches keep only the rows where both conditions are true — use whichever one makes more sense to you. Note that `Date.Year` is a number, so you don't need quotation marks around 2011.

Submission

Submit your R script (named `AssignmentName_FirstName-LastName.R`) to Canvas.

Your R script should:

- 1) Include commands and functions that are necessary to address all the questions in the assignment.
- 2) Contain comments that answer the questions in the assignment.

3) Run in its entirety without errors.

To ensure that your R script runs without errors, you should:

- Save your script.
- Navigate back to Your Workspace on Posit Cloud.
- Reopen your project.
- Run the entire script line-by-line without editing it to ensure there are no errors.

! Important

These standards apply to all submissions in this course that require R scripts. You should follow these instructions for preparation, naming, and saving of your R script for all of your individual lab assignments.